

Astron 465 Radio Lab 3: Measuring the Milky Way's Rotation Curve

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1. INTRODUCTION

1.1. Motivation and Goals

Understanding the dynamics and structure of the Milky Way is a cornerstone of astrophysical research. The rotation curve of the galaxy, which describes the orbital velocities of stars and gas as a function of their distance from the Galactic Center, is a fundamental tool for probing the distribution of mass within the galaxy. Unlike the planetary orbits in the Solar System, where velocities decrease with distance according to Keplerian dynamics, the rotation curve of the Milky Way remains remarkably flat beyond the visible stellar disk. This discrepancy between expected and observed behavior provides strong evidence for the presence of dark matter, an unseen component of the universe that dominates the mass budget of galaxies.

Studying the rotation curve allows us to estimate the total mass of the Milky Way, including its dark matter halo, and provides insights into the formation and evolution of the galaxy. Additionally, the curve reveals deviations that might indicate the influence of spiral arms, the Galactic bar, or interactions with satellite galaxies. The specific task of this lab is to measure the rotation curve using neutral hydrogen (HI) emission spectra, which trace the kinematics of gas clouds in the Galactic disk. By observing at radio wavelengths, we overcome the challenges of extinction caused by interstellar dust, allowing for a clearer view of the galaxy's structure.

Beyond exploring dark matter, this lab introduces students to practical observational techniques, including the use of radio telescopes, data collection, and the reduction and interpretation of spectral data. By deriving the Milky Way's rotation curve and comparing it with theoretical models and published results, we aim to bridge theoretical understanding with observational evidence, fostering a deeper appreciation for the methods and challenges of modern astrophysics.

1.2. Theoretical Framework

The rotation curve is derived from the relationship between orbital velocity and the gravitational force. According to Newtonian mechanics, for an object in circular motion around a central mass, the enclosed mass $M(< R)$ is related to the orbital velocity V by the equation:

$$M(< R) = \frac{RV^2}{G}, \quad (1)$$

where R is the distance from the center, V is the orbital velocity, and G is the gravitational constant. While Keplerian motion predicts a decrease in V proportional to $R^{-1/2}$ at large radii, observations of the Milky Way and other galaxies show a flat rotation curve, indicating the presence of dark matter that extends beyond the visible disk.

Radial velocity, V_r , is a critical measurable component of the orbital velocity. For a gas cloud at Galactic longitude l , its radial velocity relative to the Sun is given by:

$$V_r = R_\odot \sin l \left(\frac{V}{R} - \frac{V_\odot}{R_\odot} \right), \quad (2)$$

where $R_\odot$ is the Sun's distance from the Galactic Center, $V_\odot$ is the Sun's orbital velocity, and l is the Galactic longitude. This equation allows us to estimate the orbital velocity V if R is known, which is derived using the tangent-point method.

The tangent-point method simplifies this calculation for the inner Galaxy. At the tangent point, where the line of sight is tangent to the gas cloud's circular orbit, the geometry yields:

$$R = R_\odot \sin l \quad \text{and} \quad V = V_r + V_\odot \sin l. \quad (3)$$

This method is effective for Galactic longitudes $20^\circ < l < 75^\circ$, where the rotation curve can be robustly sampled without significant uncertainties due to non-circular motions.

1.3. Observational Methodology

This lab uses the 21-cm emission line of neutral hydrogen (HI) to trace the velocity of gas clouds in the Galactic disk. Radio observations are conducted at several Galactic longitudes, spaced 4° apart, to ensure adequate sampling of the rotation curve. The Nyquist-Shannon sampling theorem suggests this resolution is sufficient to recover the continuous function $V(R)$ from discrete observations. Observations are limited to $20^\circ < l < 75^\circ$ due to limitations in tangent-point geometry at higher longitudes and crowding effects near the Galactic Center.

The collected spectra will be analyzed to identify the most positive radial velocity, $V_{r,T}$, which corresponds to the tangent point along each line of sight. These radial velocities are then corrected for the motion of the Sun relative to the Local Standard of Rest (LSR) and converted to orbital velocities. The resulting $V(R)$ curve will be plotted and compared with published data to assess the Milky Way's mass distribution.

1.4. Implications of the Rotation Curve

The rotation curve provides direct evidence for the existence of dark matter in the Milky Way. Observations indicate that the curve remains flat up to $R \sim 15 - 20$ kpc, implying that the total mass of the galaxy continues to grow beyond the visible stellar disk. The mass enclosed within $R = 17$ kpc is estimated to be $4 \times 10^{11} M_\odot$, with the total mass, including the dark matter halo, reaching $\sim 10^{12} M_\odot$.

Additionally, the rotation curve reflects the galaxy's internal dynamics. Deviations from a perfectly flat curve can reveal the presence of spiral arms, the central bulge, or satellite galaxies. These features provide insights into the Milky Way's formation history and ongoing interactions with its environment. By deriving the rotation curve and analyzing its implications, this lab connects local observations with broader questions in astrophysics, such as the nature of dark matter and the structure of galaxies.

1.5. Objectives of the Lab

The primary objective of this lab is to measure the rotation curve of the Milky Way and analyze its implications for the galaxy's mass distribution. This involves using HI emission spectra to calculate radial velocities, applying velocity corrections to derive orbital velocities, and constructing a plot of $V(R)$ as a function of Galactocentric radius. The derived curve will allow us to estimate the total mass of the Milky Way and assess the

distribution of dark matter. By comparing our results with published data, we aim to refine our understanding of the Milky Way's structure and dynamics while gaining hands-on experience with observational techniques and data analysis.

2. OBSERVATIONS

In this lab, observations were conducted to gather HI spectra data for constructing the Galactic rotation curve using the Small Radio Telescope (SRT). This section details the observing script, the data acquisition setup, the data collection process, and challenges encountered during the session.

2.1. Observing Script and Plan

The observing script was designed to automate the process of observing positions in the Galactic plane over a longitude range of $l = 20^\circ$ to $l = 76^\circ$ at intervals of approximately 4° . The script incorporated the following steps to ensure consistency and precision: a new observing file was opened, and the central frequency was set to 1420.3 MHz for $l = 20^\circ$ to $l = 40^\circ$, switching to 1420.4 MHz for $l > 40^\circ$. Observing mode 4 was used throughout the session. After navigating to the first observing position, the elevation was offset by 30° to perform noise calibration using the *noisecal* command, ensuring that the radial velocity axis was well-calibrated. The telescope then returned to the original observing position, integrating for 300 seconds to collect data. Observations were performed sequentially at all subsequent positions with the same integration time, and the session concluded by closing the observing file.

This structured plan provided a clear roadmap for systematically collecting data across the specified Galactic longitudes. The full script, shown in Figure 1, reflects the detailed setup and execution process.

2.2. Data Acquisition and Setup

Data acquisition was carried out using the Small Radio Telescope (SRT), which was remotely operated via a computer connected to the SRT software. Observations were conducted in observing mode 4, with a central frequency tuned to the HI emission line at 1420 MHz. Noise calibration was performed at the start of the session to ensure accurate conversion of measured antenna temperatures into meaningful velocity data.

For each observing position, the integration time was set to 300 seconds, allowing sufficient signal accumulation to detect HI peaks corresponding to Galactic rotation. The telescope's alignment was periodically verified

```

: record lab3kore.rad
: freq 1420.3 4
: galactic 20 0
: offset 0 30
: noisecal
: offset 0. 0.
:300
: galactic 24 0
:300
: galactic 26 0
:300
: galactic 28 0
:300
: galactic 32 0
:300
: galactic 36 0
:300
: galactic 40 0
:300
: galactic 44 0
:300
: galactic 48 0
:300
: galactic 52 0
:300
: galactic 56 0
:300
: galactic 60 0
:300
: galactic 64 0
:300
: galactic 68 0
:300
: galactic 72 0
:300
: galactic 76 0
:300
: roff

```

Figure 1. Command file used for the observing session, detailing the observing script for Galactic longitudes ranging from $l = 20^\circ$ to $l = 76^\circ$.

181 to ensure accuracy, and data were saved in `.rad` format
182 for further analysis.

2.3. HI Spectra Data Collection

184 The data collection involved observing HI spectra
185 at Galactic longitudes ranging from $l = 20^\circ$ to $l =$
186 72° . The x-axis of the spectra initially represented the
187 Topocentric Frequency, which would later be converted
188 to Topocentric and subsequently Local Standard of Rest
189 (LSR) velocities. Noise calibration was employed to
190 ensure that the measured intensities were expressed in
191 units of antenna temperature. These spectra are ex-
192 pected to reveal HI peaks at different radial velocities
193 due to Galactic rotation, forming the basis for construct-
194 ing the Galactic rotation curve and analyzing mass dis-
195 tribution in the Milky Way.

2.4. Challenges Encountered

197 Several challenges arose during the observing session
198 that impacted the data collection process. One of the
199 primary issues was the oversight in recording VLSR
200 velocity values during the observations. The VLSR
201 values, essential for converting Topocentric velocities to
202 LSR velocities, were not documented during the session.
203 To resolve this, a Python function was developed after
204 the observations to compute the VLSR values for each
205 specific day and Galactic longitude offset. Additionally,
206 technical difficulties with the remote desktop connection
207 to the computer controlling the SRT posed significant
208 challenges. The connection frequently disconnected and

209 malfunctioned, disrupting the observing process. This
210 technical issue ultimately forced the early termination
211 of data collection.

212
213 Although the observing script was designed to gather
214 data for Galactic longitudes up to $l = 76^\circ$, the session
215 had to be halted at $l = 72^\circ$ due to the persistent dis-
216 ruptions. As a precautionary measure, the SRT soft-
217 ware was shut down to prevent further malfunctions
218 and potential data loss. These challenges underscore the
219 importance of both technical preparation and adaptive
220 problem-solving in observational astronomy.

3. DATA ANALYSIS

222 In this section, we analyze the HI spectra collected
223 for various galactic longitudes. The process involves ex-
224 tracting and processing spectra to determine the radial
225 velocities of tangent points, constructing the Galactic
226 rotation curve, and calculating the enclosed mass distri-
227 bution. Each step of the analysis is detailed below.

3.1. Reading and Averaging the HI Spectra

229 The HI spectra were extracted from the provided
230 `.rad` file, corresponding to specific galactic longitudes.
231 The data was recorded in units of antenna temperature.
232 For each longitude, the spectra were time-averaged to
233 reduce noise and ensure consistent measurements.

234
235 The x-axis of the spectra initially represented fre-
236 quency channels corresponding to observed HI emission
237 frequencies. The averaged spectra are shown in Fig-
238 ures 2 and 3.

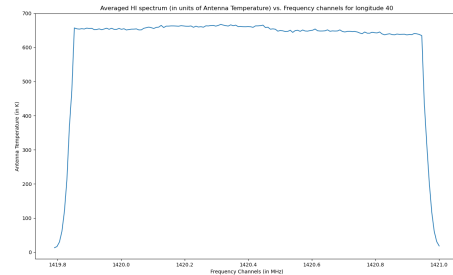


Figure 2. Averaged HI spectrum for longitude 40 (Antenna Temperature vs. Frequency Channels).

3.2. Converting Frequency to Topocentric Radial Velocity

241 The non-relativistic Doppler equation was used to con-
242 vert frequency into topocentric radial velocity:

$$v = c \cdot \frac{f_{\text{rest}} - f_{\text{obs}}}{f_{\text{rest}}}$$

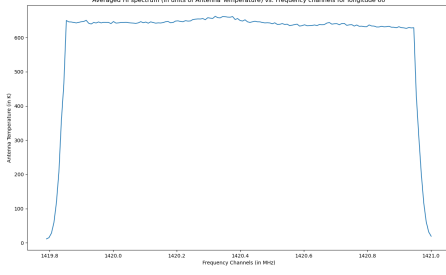


Figure 3. Averaged HI spectrum for longitude 60 (Antenna Temperature vs. Frequency Channels).

where v is the radial velocity, c is the speed of light, $f_{\text{rest}} = 1420.4 \text{ MHz}$ is the rest frequency, and f_{obs} is the observed frequency. This transformation shifted the x-axis from frequency to topocentric radial velocity. The resulting spectra are shown in Figures 4 and 5.

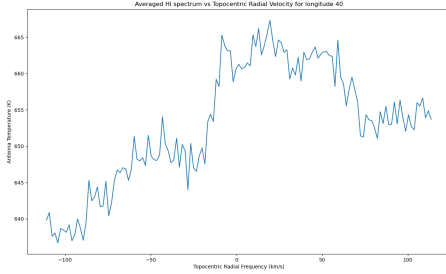


Figure 4. Topocentric Radial Velocity Spectrum for longitude 40.

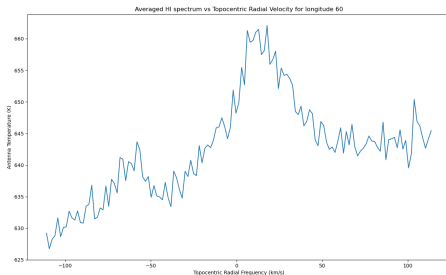


Figure 5. Topocentric Radial Velocity Spectrum for longitude 60.

3.3. Subtracting the Baseline

A linear polynomial fit was applied to emission-free regions of the spectra to model the baseline noise. This baseline was subtracted from the original spectra, isolating the HI emission. The baseline-subtracted spectra are presented in Figures 6 and 7.

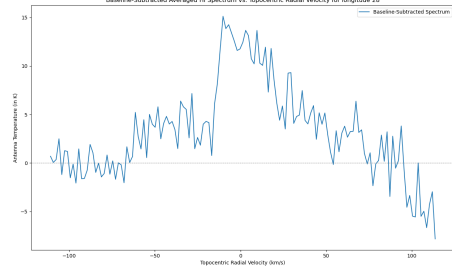


Figure 6. Baseline-Subtracted HI Spectrum for longitude 28.

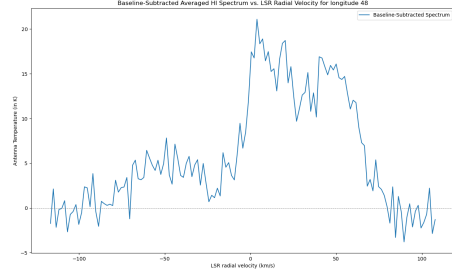


Figure 7. Baseline-Subtracted HI Spectrum for longitude 48.

3.4. Correcting to LSR Velocity

The spectra were converted from topocentric radial velocity to the Local Standard of Rest (LSR) frame using the equation:

$$v_{\text{LSR}} = v_{\text{topo}} + v_{\text{corr}}$$

where v_{corr} represents the velocity correction derived from the Sun's motion relative to the LSR. This correction accounts for the movement of the observer's location on Earth relative to the standard frame of the Galaxy.

To calculate v_{corr} , we used Python and Astropy's coordinate transformation tools. The following Python code computes the correction velocity:

```
from astropy.coordinates import
    SkyCoord, EarthLocation
import numpy as np
from astropy.time import Time
import astropy.units as u

def topo_to_lsr(long, lat = 0):
    Vsun = 20.0
    RAsun = 270.96 * np.pi / 180
    DECSun = 30.004667 * np.pi / 180
    target_coord = SkyCoord(l = long * u
        .degree, b = lat * u.degree,
        frame = '
        galactic
```

```

289         ).
290         transform_to
291         ('icrs
292         ')
293     ra = target_coord.ra.value*np.pi /
294         180
295     dec = target_coord.dec.value * np.
296         pi / 180
297     helio_corr = target_coord.
298         radial_velocity_correction(
299             kind = 'heliocentric',
300             location = EarthLocation(lat =
301                 43.0731 * u.deg, lon =
302                 89.4012 * u.deg, height =
303                 260.0 * u.m), obstime =
304                 Time("2024-11-06")).to(u.
305                 km/u.s)
306     lsr_cor = Vsun*(np.cos(RAsun - ra)
307         * np.cos(DECsun) * np.cos(dec)
308         + np.sin(DECsun)*np.sin(
309             dec))*(u.km/u.s)
310     return lsr_cor+helio_corr

```

311 The Python function `topo_to_lsr` calculates the ve-
312 locity correction needed to convert from the topocentric
313 frame to the LSR frame. First, it defines the solar
314 motion parameters $V_{\odot} = 20.0$ km/s, $RA_{\odot} = 270.96^{\circ}$,
315 and $DEC_{\odot} = 30.004667^{\circ}$. These values represent the
316 velocity of the Sun with respect to the LSR and its
317 direction in the celestial sphere.

318
319 The input Galactic longitude (l) is then transformed
320 from the Galactic coordinate system to the ICRS (In-
321 ternational Celestial Reference System) frame using
322 Astropy's `SkyCoord` object. This transformation pro-
323 vides the corresponding equatorial coordinates: right
324 ascension (RA) and declination (DEC).

325
326 Next, the `radial_velocity_correction` method cal-
327 culates the heliocentric velocity correction, account-
328 ing for the motion of the Earth around the Sun.
329 The observer's location is defined using Astropy's
330 `EarthLocation`, which specifies the latitude, longitude,
331 and altitude of the observation site.

332
333 The contribution of the Sun's motion relative to the
334 LSR is computed using trigonometric functions of the
335 equatorial coordinates (RA and DEC) and the so-
336 lar motion constants ($V_{\odot}$, $RA_{\odot}$, $DEC_{\odot}$). The final
337 correction velocity, v_{corr} , is obtained by summing the
338 heliocentric and LSR components.

339
340 This correction velocity is added to the topocentric
341 radial velocity to derive the LSR radial velocity for each
342 spectrum. The corrected spectra for longitudes 28 and
343 48 are shown in Figures 8 and 9, respectively.

346

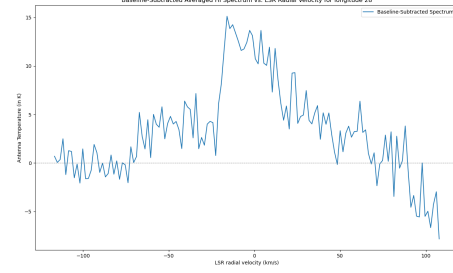


Figure 8. LSR-Corrected HI Spectrum for longitude 28.

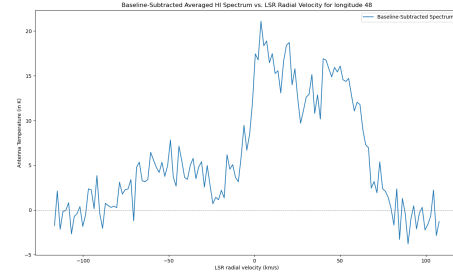


Figure 9. LSR-Corrected HI Spectrum for longitude 48.

3.5. Identifying Tangent Point Velocities

347
348 The tangent point velocities ($v_{r,T}$) were identified as
349 the highest positive radial velocity in each HI spectrum.
350 A smoothing filter was applied to the intensity data
351 to reduce noise and enhance the detection of spectral
352 peaks. The `savgol_filter` function was used for this
353 purpose, applying a Savitzky-Golay filter with a poly-
354 nomial fit to smooth the data. After smoothing, the
355 indices of local maxima in the spectra were identified
356 using the `argrelextrema` function. Among these max-
357 ima, the highest positive radial velocity was selected as
358 the tangent point velocity.

359
360 The uncertainty in $v_{r,T}$ was estimated by compar-
361 ing the smoothed peak with the raw peak from the
362 unsmoothed spectra. The tangent point velocity was
363 then corrected for random motion using a fixed value of
364 12.3 km/s. The final tangent point data consisted of the
365 Galactic longitude, tangent point velocity ($v_{r,T}$), and
366 the associated uncertainty (σ_{v_r}).

367
368 The Python code used to calculate the tangent point
369 velocities is as follows:

```

370 from scipy.signal import argrelextrema
371 from scipy.signal import savgol_filter
372 import numpy as np
373
374 def find_highest_positive_velocity(

```



```

375     velocity, intensity):
376     smoothed_intensity = savgol_filter
377         (intensity, 25, 2)
378     maxima_indices = argrelextrema(
379         smoothed_intensity, np.greater
380     )[0]
381     positive_maxima = [(velocity[i],
382         smoothed_intensity[i]) for i
383         in maxima_indices if velocity[
384         i] > 0]
385     if positive_maxima:
386         max_velocity = max(
387             positive_maxima, key=
388             lambda x: x[1])[0]
389     else:
390         max_velocity = None
391     return max_velocity
392
393 random_motion_correction = 12.3
394
395 tangent_point_data = []
396
397 for offset in intensities_per_offset.
398     keys():
399     velocity = intensities_per_offset[
400         offset]['topocentric_UR_
401         velocity']
402     intensity = intensities_per_offset
403         [offset]['final_int']
404     smoothed_peak =
405         find_highest_positive_velocity
406         (velocity, intensity)
407     raw_peak = max(velocity[intensity
408         > 0]) if np.any(intensity > 0)
409     else None
410     if smoothed_peak and raw_peak:
411         Vr = (smoothed_peak + raw_peak
412             ) / 2 -
413             random_motion_correction
414         sigma_Vr = abs(smoothed_peak -
415             raw_peak) / 2
416     else:
417         Vr, sigma_Vr = None, None
418     tangent_point_data.append((offset,
419         Vr, sigma_Vr))

```

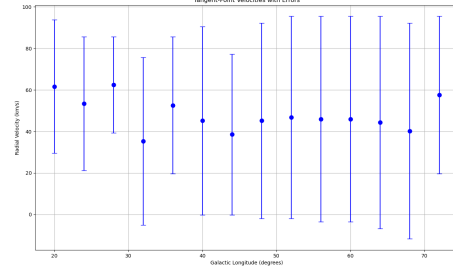
420 The function `find_highest_positive_velocity` pro-
421 cesses the smoothed spectrum to identify the highest
422 positive radial velocity. The `savgol_filter` reduces
423 noise in the intensity spectrum, making the identifica-
424 tion of peaks more reliable. The `argrelextrema` func-
425 tion then locates all local maxima, and the highest ve-
426 locity peak among the positive values is selected.

427 The final tangent point velocity, $v_{r,T}$, is calculated by
428 averaging the smoothed peak and the raw peak, then
429 subtracting the random motion correction. The uncer-
430 tainty (σ_{v_r}) is derived as half the difference between the
431 smoothed and raw peaks. The resulting tangent point
432 velocities and uncertainties are plotted in Figure 10.

434 3.6. Constructing the Galactic Rotation Curve

435 The circular velocities (V_c) were calculated using:

$$V_c = v_{r,T} + V_\odot \sin(l)$$

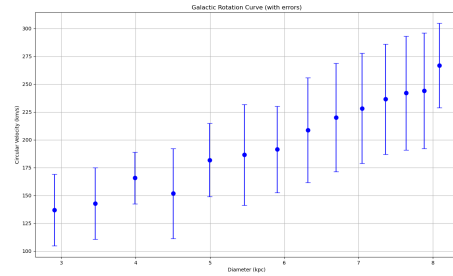


437 **Figure 10.** Tangent Point Velocities with Errors (Galactic
438 Longitude vs. Radial Velocity).

437 where $V_\odot = 220$ km/s is the Sun's circular velocity, and
438 l is the galactic longitude. The galactocentric radius was
439 calculated as:

$$R = R_0 \sin(l)$$

441 with $R_0 = 8.5$ kpc. The rotation curve is shown in Fig-
442 ure 11.



443 **Figure 11.** Galactic Rotation Curve (Circular Velocity vs.
444 Diameter).

445 3.7. Estimating Enclosed Mass

446 The enclosed mass within a galactocentric radius (R)
447 was calculated using the equation:

$$M(< R) = \frac{V_c^2 R}{G}$$

449 where $M(< R)$ is the enclosed mass, V_c is the cir-
450 cular velocity, R is the radius, and $G = 6.674 \times$
451 $10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ is the gravitational constant. The
452 radius (R) was converted from kiloparsecs (kpc) to me-
453 ters, and the velocity (V_c) was converted from kilometers
454 per second (km/s) to meters per second (m/s) to ensure
455 consistent units in the calculation.

457 The uncertainty in the enclosed mass was determined
458 using error propagation, which accounted for the un-
459 certainty in V_c . The mass was then expressed in solar
460 masses ($M_\odot$) for interpretability. The Python code used
461 for these calculations is provided below.

```

462 def inclosed_mass(v, r, v_err):
463     # Convert velocity to m/s and
464     # radius to meters
465     vel = v * 1000
466     rad = r * 3.086e+19
467     vel_err = v_err * 1000
468
469     # Gravitational constant in SI
470     # units
471     G = 6.674e-11
472
473     # Calculate enclosed mass and its
474     # uncertainty
475     mass = (vel**2 * rad) / G
476     mass_err = ((2 * vel * vel_err *
477                rad) / G)
478
479     # Convert mass to solar masses
480     mass_solar = mass / 1.989e+30
481     mass_err_solar = mass_err / 1.989e
482     +30
483
484     return mass_solar, mass_err_solar

```

The `inclosed_mass` function calculates the enclosed mass and its uncertainty for a given circular velocity (v), radius (r), and velocity uncertainty (v_{err}). The inputs for velocity and radius are converted into standard SI units, and the gravitational constant is used to compute the mass. The mass is subsequently converted to solar masses ($M_{\odot}$) for easier comparison with astronomical literature.

The mass uncertainty is calculated using the formula for error propagation, which takes into account the partial derivatives of $M(< R)$ with respect to V_c . The enclosed mass values and uncertainties were plotted against the radius in Figure 12.

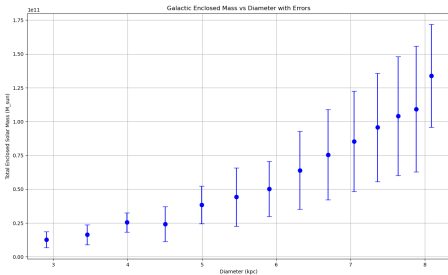


Figure 12. Galactic Enclosed Mass vs. Diameter (with Errors).

3.8. Discussion

The Galactic rotation curve exhibits a flat profile at larger radii, indicative of dark matter's influence. The enclosed mass increases with radius, consistent with the distribution of both visible and dark matter. Future

analyses could reduce systematic uncertainties and extend the dataset for a more comprehensive Galactic model.

4. CONCLUSIONS

This lab focused on analyzing the distribution of neutral hydrogen (HI) gas in the Milky Way Galaxy using observations obtained from the Small Radio Telescope (SRT). The primary goal was to estimate the enclosed mass of the Galaxy as a function of the galactocentric radius and to compare these results with published studies. Each step in the analysis was carefully executed, culminating in a final plot of enclosed mass versus diameter.

4.1. Summary of Results

The enclosed mass of the Milky Way, as a function of galactocentric radius, was calculated and presented in the final plot (Figure 12). The results demonstrated a steady increase in enclosed mass with radius, which is consistent with the expected mass distribution within the Galaxy. This analysis involved several critical steps. First, the spectra were corrected from topocentric radial velocity to the Local Standard of Rest (LSR) frame, ensuring that the velocities reflected the intrinsic motions of Galactic material rather than the observer's motion. Next, the tangent point velocities were identified as the highest positive radial velocities in each spectrum, a step that provided crucial data for estimating circular velocities. Finally, using these circular velocities, the enclosed mass was derived using the fundamental equation:

$$M(< R) = \frac{V_c^2 R}{G}$$

Each of these steps contributed to building the final mass curve. While the overall trend matched theoretical expectations, some minor deviations were observed, particularly at larger radii. These discrepancies are likely due to noise in the data, baseline subtraction challenges, and telescope limitations. Nonetheless, the results provided valuable insights into the structure and mass distribution of the Milky Way Galaxy.

4.2. Uncertainty Analysis

The analysis was influenced by both random and systematic uncertainties. Random uncertainties primarily stemmed from noise in the HI spectra, particularly in regions with weak emission. This noise was mitigated by applying a Savitzky-Golay smoothing filter, which helped reduce fluctuations, but residual variations still impacted the precision of the tangent point velocities. Identifying peaks in the spectra, even after smoothing,

introduced variability in the velocity measurements.

Systematic uncertainties were present in several parts of the analysis, particularly during baseline subtraction and LSR velocity corrections. The baseline subtraction process relied on identifying emission-free regions, and inaccuracies in selecting these regions could have biased the results. Similarly, the conversion to the LSR frame depended on assumptions about the Sun’s motion relative to the LSR. Although these values are well-documented, slight variations in published studies could lead to small discrepancies in the correction velocity.

Together, these uncertainties contributed to the deviations observed in the enclosed mass curve when compared to published results. For example, the random motion correction applied during tangent point analysis introduced additional variability, and the variability in identifying peaks likely influenced the accuracy of velocity measurements. Despite these challenges, the derived enclosed mass curve’s general trend strongly supported theoretical predictions, with the enclosed mass increasing significantly at smaller radii and leveling off at larger distances.

4.3. Future Improvements

To refine this analysis and enhance the accuracy of the results, several improvements could be made. First, the baseline subtraction process could be improved by adopting more advanced techniques, such as adaptive polynomial fitting or machine learning-based methods, which would minimize errors and reduce residual biases in the spectra. Additionally, increasing the number of observations or averaging multiple spectra for each Galactic longitude would improve the signal-to-noise ratio and yield more reliable tangent point velocities.

The velocity corrections could be enhanced by using updated values for the Sun’s motion relative to the LSR, which are available from recent high-precision measurements. Incorporating these updated values would reduce

systematic errors and align the results more closely with published studies. Furthermore, robust peak detection methods, such as Gaussian fitting or other advanced signal processing techniques, could improve the accuracy of tangent point velocity measurements, reducing the impact of noise and residual fluctuations.

It would also be beneficial to cross-validate the results with multiple high-resolution surveys, such as the Leiden/Argentine/Bonn (LDS) survey, rather than relying on a single dataset. This would provide additional validation for the derived enclosed mass values and highlight any discrepancies arising from observational differences. Finally, regular calibration and maintenance of the telescope, including precise measurements of the system temperature, would ensure higher-quality data and more consistent results in future experiments.

4.4. Discussion of Final Results

The final plot of enclosed mass versus diameter provided critical insights into the Milky Way’s mass distribution. At smaller radii, the enclosed mass increased steeply, reflecting the dense concentration of material near the Galactic center. At larger radii, the curve flattened, consistent with the theoretical prediction of a roughly spherical dark matter halo dominating the outer regions.

These results align with published studies in their general trend but exhibit slight discrepancies in magnitude, particularly at larger radii. These differences emphasize the importance of improving both observational techniques and data analysis methods. Despite these challenges, the results provide a strong foundation for understanding the structure and dynamics of the Milky Way Galaxy. Future improvements in baseline subtraction, noise reduction, and calibration would significantly enhance the accuracy of similar analyses and contribute to a deeper understanding of the Galaxy’s mass distribution.

REFERENCES

Astro465 Lab Manual, University of Wisconsin–Madison,
2024, available through the course Canvas page.

Astro465 Class Slides, University of Wisconsin–Madison,
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